

4(a)(i)	$m = \rho \times V$	C1
	$= (4 / 3) \times \pi \times (1.2 \times 10^{-6})^3 \times 940 = 6.8 \times 10^{-15} \text{ (kg)}$	A1
4(a)(ii)	minimum charge (on drop) is $1.6 \times 10^{-19} \text{ C}$	B1
4(b)(i)	$V = Ed$	C1
	$V = 2.1 \times 10^5 \times 8.0 \times 10^{-3}$	A1
	$= 1.7 \times 10^3 \text{ V}$	
4(b)(ii)	constant velocity so no resultant force (so in equilibrium)	B1
4(b)(iii)	$mg = Eq$ or $mg = (V / d)q$ or $F = mg$ and $F = Eq$	C1
	$q = (6.8 \times 10^{-15} \times 9.81) / 2.1 \times 10^5$ $= 3.2 \times 10^{-19} \text{ C}$	A1
	sign of charge is negative	A1
4(c)(i)	<u>electric</u> force decreases	B1
	weight > electric force or resultant force acts downwards	B1
4(c)(ii)	(field line) separation increases	B1
4(d)(i)	upthrust (force)	B1
4(d)(ii)	air resistance/drag/viscous (force)	B1